

the grid, that do not have access to any form of power or wires or distribution lines. In those areas Gram Power has to set up the entire micro grid with its own renewable power generation, distribution infrastructure, etc. and they end up supplying single phase power to every single consumer just like we receive in our homes in urban areas. In these areas, for instance, if in the future the national grid were to arrive, then since Gram Power's distributional infrastructure is already distributing the standard single-phase power, the grid basically becomes another source of power in that microgrid.

"Whenever the grid integrates with our system, we'll just sync the phase with a step down transformer, because we're supplying AC power on our line and there will also be AC power coming from the grid, so you just have to sync both the supplies," says Yashraj. "And whenever grid power is available, you have grid power on the lines, whenever grid power is not available, you have your renewable generation or you have your backup power from the batteries getting into that local micro grid. So that is making power actually reliable in these off-grid areas."

Challenges in rolling out Gram Power

Volatility in the electricity market in India is something that Gram Power had a major headache in dealing with. Yashraj explains, "Well, one challenge is of course that the market changes very quickly, so when we were in college, at that point the energy situation in India was quite different. To give you an example, when I started research on this project, at that point we were working in purely off grid areas, and we were only working on this simple smart battery that people could use to run a whole range of low-power appliances like DC fans, and some small TVs and lights and things like that.

"But by the time we completed that product, came and launched it here in India, the villages in those two-three years that we were doing research, had already been electrified. So since all that electrification does not equate to access to power, because having a wire in your house in no way means that you're actually getting power supply as well."

But this led to a unique problem, even a desirable problem, Yashraj

explains. People's needs for power started getting converted into demand. Instead of being content with having electricity at all, people wanted access to electricity for much larger appliances. Yashraj states that this was a great thing to witness, spurring Gram Power on to electrify rural India with much gusto. "I think what was challenging was to make sure that what you're developing is something that can really have a strong fundamental impact in the sector that you're working in, and it is something that is bringing in much more sustainability like a lot of our technology development is focused on how to make power distribution much more sustainable in rural India, which today is completely loss-making sector for all power companies," feels Yashraj.

Then there were business challenges, since the market for off-grid energy in India is so big, it also attracts a whole lot of shoddy suppliers and shoddy products out there, claims Yashraj. "So initially it was a big challenge for us because we burned our fingers with a lot of shoddy suppliers over equipment. And that was a major challenge – the subsidy environment. The regulatory environment in the country is so ambiguous and the subsidy environment keeps changing. Every year the government comes up with a new policy which makes private investment quite difficult. That was one of the main reasons why we started moving from the government funded, government subsidized solar micro grid to the pure smart grid module. We're moving now to a business module where it's completely unsubsidised, like we don't require subsidy from the government to make our business model and our technology sustainable."

Technology powering Gram Power powering Social Change?

Yashraj isn't coy about the pivotal role that modern technology has played in the development and deployment of the Gram Power electricity model across 17 off-the-grid rural villages in the country. Technology is definitely an enabler for them, becoming a key USP that differentiates Gram Power from the existing power grid operational in the country.

"Technology is actually helping us solve almost all the problems that we identify," says Yashraj, "like the payments system that we've developed is possible only



YASHRAJ KHAITAN

Yashraj Khaitan is the co-founder and CEO of Gram Power, an energy technology company out of University of California - Berkeley that is working on introducing a new paradigm of power distribution in rural India. Twenty-four year old Yashraj started working on Gram Power when he was 19 and prior to that worked on cutting edge solar cell research as a high school student. He completed his B.S. in Electrical Engineering and Computer Science from the University of California - Berkeley and then opted out of his post graduate program to work on Gram Power permanently.

because of the wireless technology that we're using. All the remote monitoring, all the remote two-way communication, grid monitoring, theft detection systems, it's all been made possible because of the hardware and software that we've developed. While of course there is a strong social component to the entire thing, you can't just go ahead and put technology in the village and expect that everything will suddenly change. People have to be told of the advantages of using electricity in the right manner and how technology is enabling the entire distribution to become more sustainable, but at the end of the day, it is the technology that is allowing us to put the right kind of checks and balances in the power distribution. So technology is the enabler for everything that we're doing."

And so far, in the pilot runs that Gram Power's had across villages, rural India's becoming more efficient than city slickers